

Valid inequalities for convex cones

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A *convex cone* is a set that contains all conic combinations of its elements. A *polyhedral cone* is a convex cone that is also a polyhedron. In this note we characterize valid inequalities for convex cones. A consequence is a characterization of the inequalities needed to describe a polyhedral cone.

The valid inequalities for a convex cone are characterized as follows:

Theorem 1 (*Characterization of the valid inequalities for a convex cone*)
Let $C \subseteq \mathbb{R}^n$ be a convex cone. An inequality $\alpha^T x \leq \beta$ is valid for $C \iff$ the inequality $\alpha^T x \leq 0$ is valid for C and $\beta \geq 0$. ■

Proof: " $\implies$ ": Suppose $\alpha^T x \leq \beta$ is valid for C . One must have $\beta \geq 0$ since otherwise $\alpha^T x \leq \beta$ cuts off the origin. Furthermore, if one had $\alpha^T \bar{x} > 0$ for some $\bar{x} \in C$, then $\mu \cdot \bar{x} \in C$ and $\alpha^T(\mu \cdot \bar{x}) > \beta$ would hold for $\mu > 0$ sufficiently large, which would contradict validity of $\alpha^T x \leq \beta$ for C . Hence $\beta \geq 0$ and $\alpha^T x \leq 0$ is valid for C .

" $\impliedby$ ": If $\beta \geq 0$ and $\alpha^T x \leq 0$ is valid for C , then $\alpha^T x \leq 0 \leq \beta$ for all $x \in C$, which implies validity of $\alpha^T x \leq \beta$ for C . ■

Applying Theorem 1 directly to polyhedral cones shows that polyhedral cones are those polyhedra with the zero vector as the right-hand-side vector:

Corollary 1 (*Inequality description of a polyhedral cone*)
If $C \subseteq \mathbb{R}^n$ is a polyhedral cone, then $C = \{x \in \mathbb{R}^n : Dx \leq 0_p\}$ holds for some matrix $D \in \mathbb{R}^{p \times n}$. ■

Indeed, if $C \subseteq \mathbb{R}^n$ is a polyhedral cone, then $C = \{x \in \mathbb{R}^n : Dx \leq d\}$ holds for some $D \in \mathbb{R}^{p \times n}$ and $d \in \mathbb{R}^p$. We claim $C = \{x \in \mathbb{R}^n : Dx \leq 0_p\}$.

Since the inequalities $Dx \leq d$ are valid for C , Theorem 1 shows $d \geq 0_p$ and that the inequalities $Dx \leq 0_p$ are valid for C .

From validity of $Dx \leq 0_p$ for C , one has $C \subseteq \{x \in \mathbb{R}^n : Dx \leq 0_p\}$. Conversely, since $d \geq 0_p$, the inequalities $Dx \leq 0_p \leq d$ hold for all points in $\{x \in \mathbb{R}^n : Dx \leq 0_p\}$, and therefore $\{x \in \mathbb{R}^n : Dx \leq 0_p\} \subseteq C$.